Aayush Kumar

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Education

National Institute of Technology, Kurukshetra **Nov 2021 – May 2025**
B.Tech in Computer Engineering

Experience

Airbus — Software Developer 1 **Nov 2025 – Present**
- Build and maintain a **C++/Qt/QML** iPad application for flight-operations workflows, used daily by **40,000 pilots**, with a focus on responsiveness, stability and predictable behaviour.
- Cut main-thread memory usage by **23%** by profiling with Xcode Instruments and tightening object and resource lifetimes on hot code paths.
- Designed **state-machine-driven** workflows that guarantee deterministic, traceable state transitions in **safety-critical** scenarios.
- Integrated a fine-tuned aviation-domain small language model into the app for context-aware operational queries.

Airbus — DevOps Intern **Jun 2025 – Nov 2025**
- Built a proof of concept for migrating on-premises infrastructure to **AWS** that the team **adopted** as the basis for its migration, after evaluating scalability, deployment feasibility and operational requirements.
- Automated cloud workflows with **AWS Lambda** and **S3**, and configured **Route 53** DNS for service integration.
- Built **Docker**-based CI/CD pipelines plus monitoring scripts and APIs for deployment validation and log-based troubleshooting.

Turing — Software Developer **Nov 2024 – Jun 2025**
- Built **GitHub Actions** evaluation pipelines that scored LLM output on backend engineering, systems, competitive programming and debugging tasks.
- Built a **Dockerized reinforcement-learning evaluation environment** that compared model output against reference annotations for **Meta** and **Alibaba** projects.
- Worked with enterprise client teams to identify recurring model failure cases and debugging patterns, improving RLHF dataset workflows.

Projects

lob-engine | ***C++20, Google Benchmark, GoogleTest, sanitizers*** **Write-up** | **Source Code**
- Price-time-priority **matching engine** (pre-allocated pool, intrusive queues, O(1) cancel); flat tick-indexed levels with a bitset, then an open-addressing id table: **20.9M → 35.7M → 65.2M ops/s** on a 1M-op mixed workload.
- Replayed 20 minutes of recorded Binance BTCUSDT data (563k order ops): **top 10 levels matched the exchange** after 99.93% of depth updates; 5.4× faster than the `std::map` baseline on a small book that fits in cache.
- Verified with **differential tests** against naive reference implementations, **16 planted bugs, all caught**; CI under ASan/UBSan.

Trace Studios | ***Next.js, Flask, Raspberry Pi, Docker, Supabase, Turso***
- Co-built and launched a full-stack **photo-booth platform** with photo capture, digital sharing and Raspberry Pi thermal printing, working across the stack.
- Worked on the Flask printing pipeline and cloud-to-edge gallery sync, with **HMAC-signed uploads** and QR-based photo access.

Technical Skills

Languages: C++, C, Python, Go, Java, Bash
Systems: Linux, performance profiling, state machines, Docker, Kubernetes, CI/CD (GitHub Actions)
Cloud: AWS (Lambda, S3, Route 53), GCP, monitoring and automation
Frameworks: Qt/QML, Flask, REST APIs

Certifications

- **Certified Kubernetes Administrator (CKA)** and AWS Certified Cloud Practitioner, earned 2023 (both expired).

Leadership

- **President, SUC Dance Crew:** led a 35-member team to 7 inter-college wins. **Co-organiser, TEDxNITKKR:** ran the event end to end for 300+ attendees and 8 speakers.